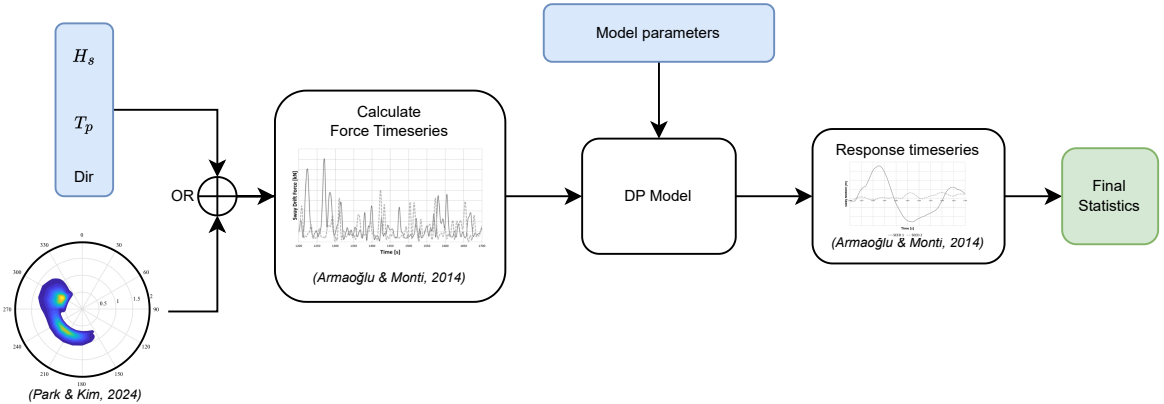
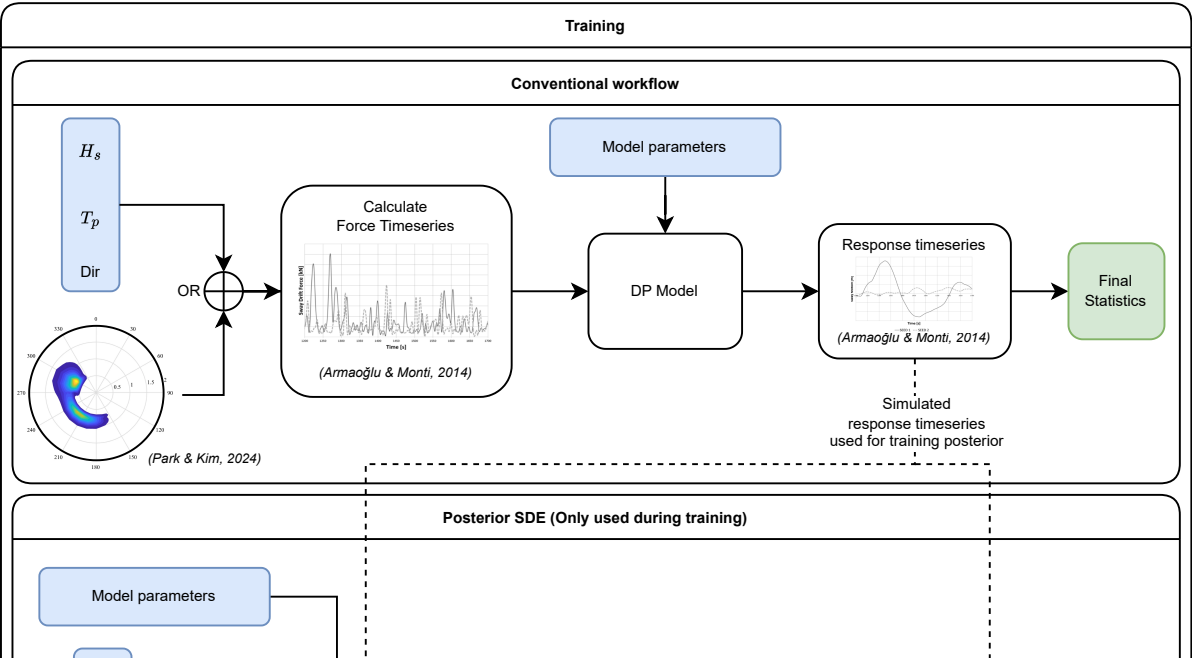
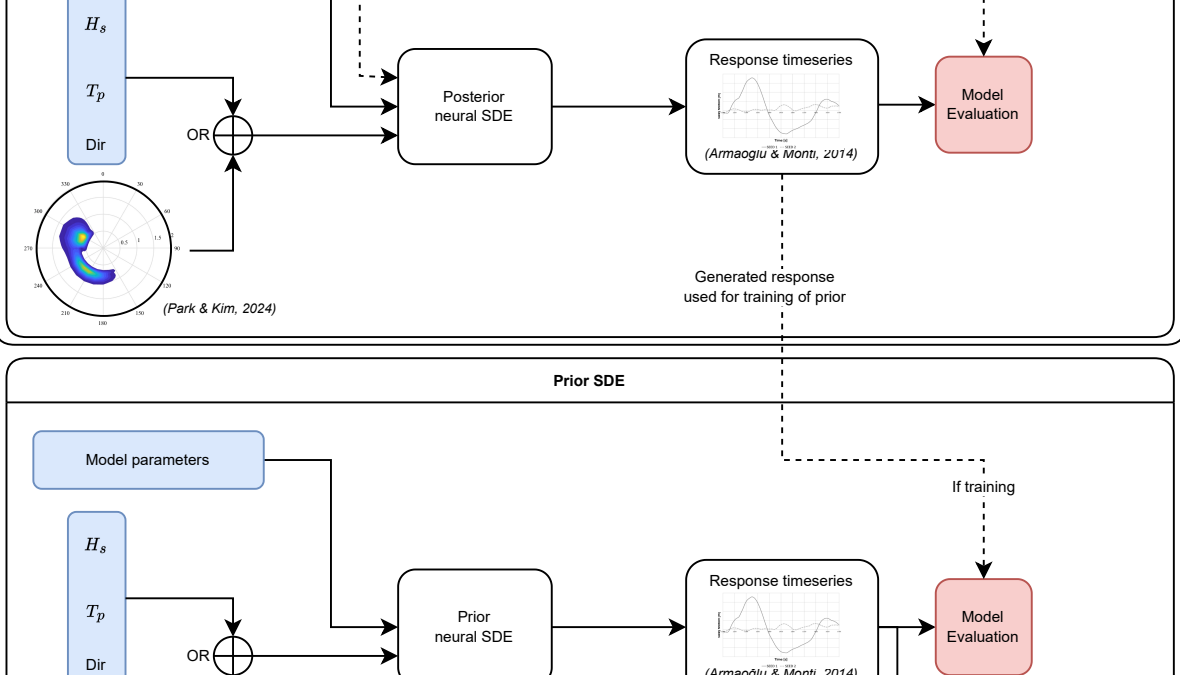
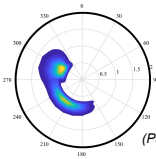

H_s
T_p
Dir
OR
(Park & Kim, 2024)
Calculate
Force Timeseries
(Armaoğlu & Monti, 2014)
Model parameters
DP Model
Response timeseries
(Armaoğlu & Monti, 2014)
Final
Statistics


Training
Conventional workflow
H_s
T_p
Dir
OR
(Park & Kim, 2024)
Calculate
Force Timeseries
(Armaoğlu & Monti, 2014)
Model parameters
DP Model
Response timeseries
(Armaoğlu & Monti, 2014)
Final
Statistics
Simulated
response timeseries
used for training posterior
Posterior SDE (Only used during training)
Model parameters

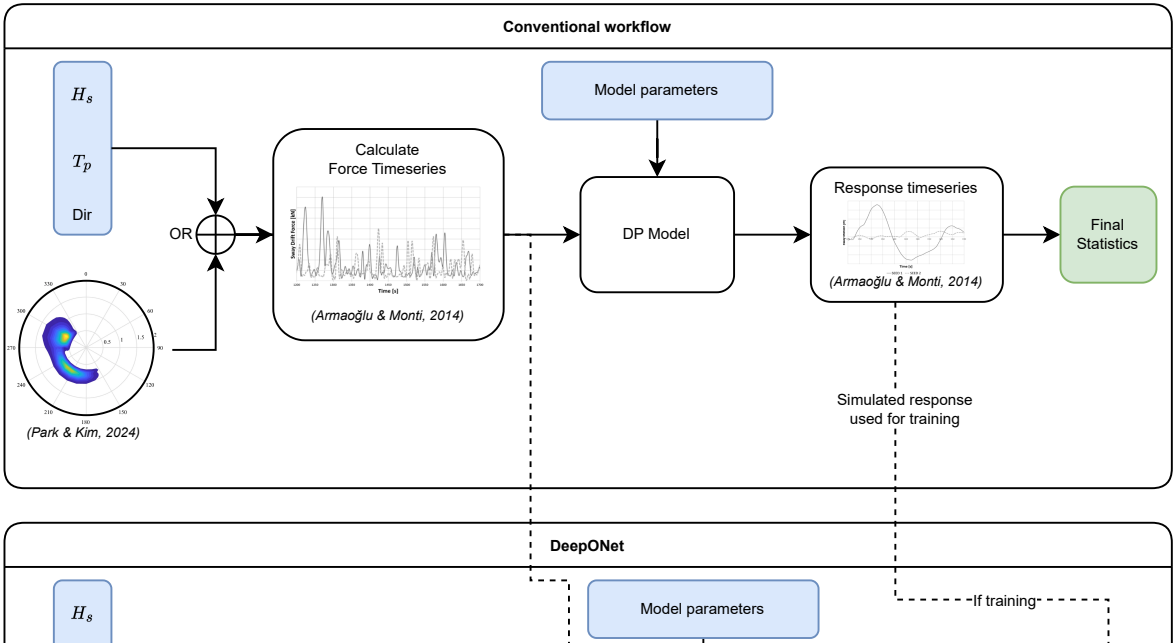

H_s
T_p
Dir
OR
Posterior neural SDE
Response timeseries
(Armaogiu & Monti, 2014)
Model Evaluation
(Park & Kim, 2024)
Generated response used for training of prior
Prior SDE
Model parameters
If training
H_s
T_p
Dir
OR
Prior neural SDE
Response timeseries
(Armaogiu & Monti, 2014)
Model Evaluation

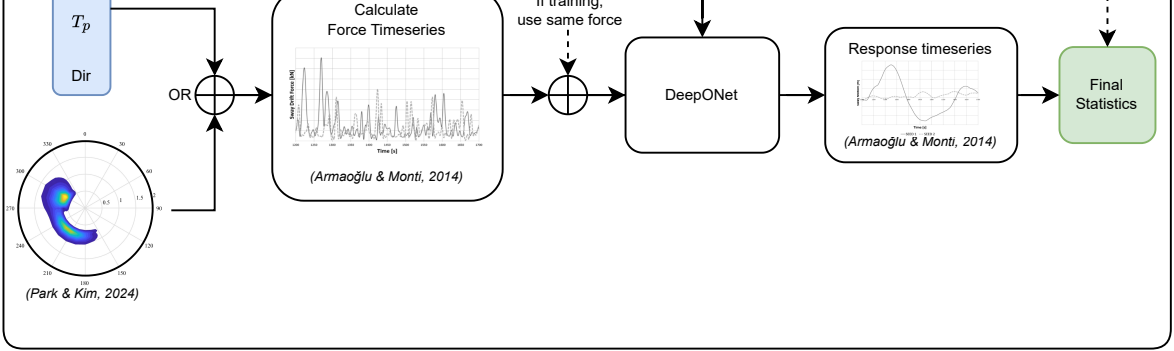


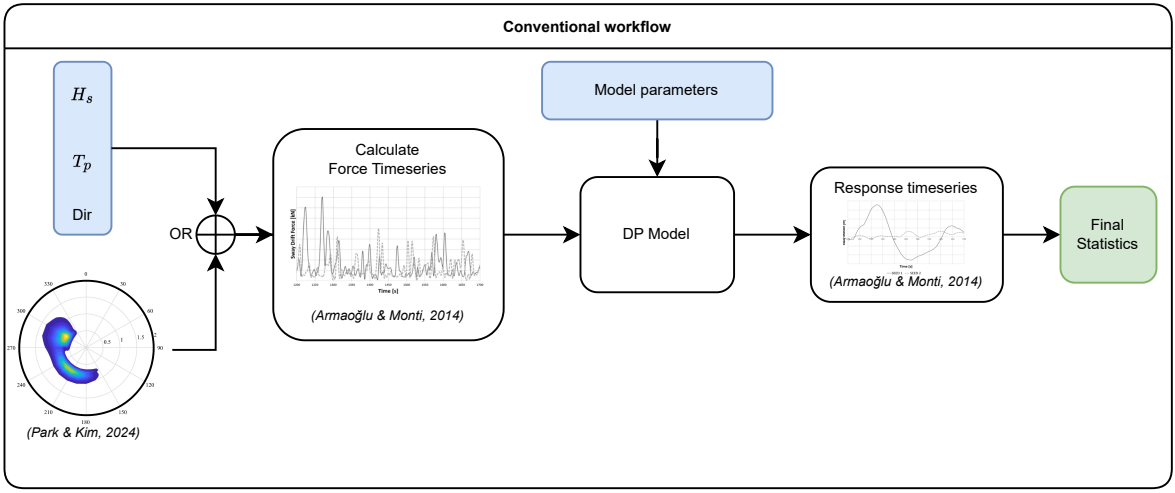
(Park & Kim, 2024)

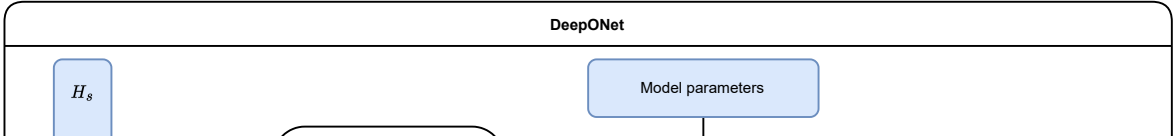
(Armagio & Monti, 2014)

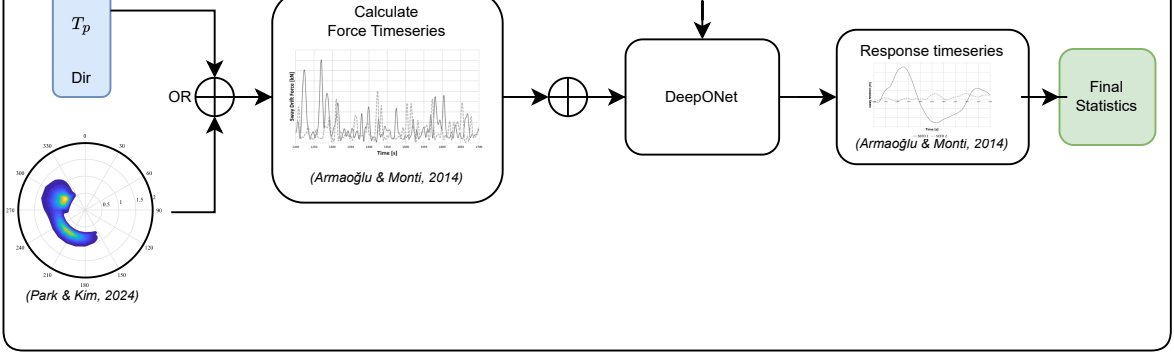
Final Statistics

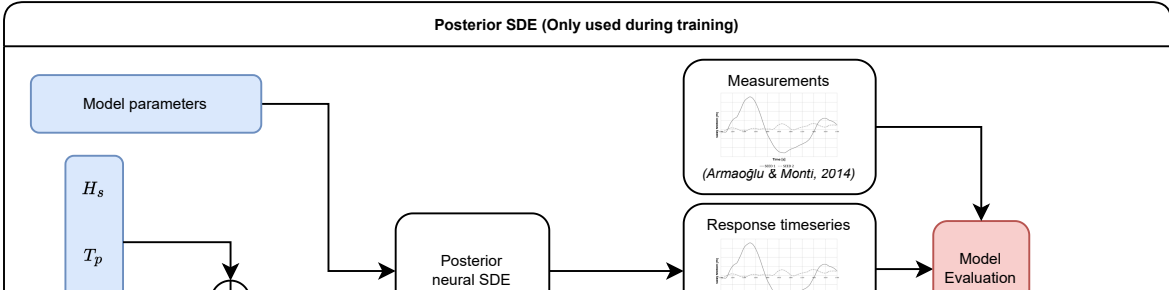

Conventional workflow
H_s
T_p
Dir
OR
(Park & Kim, 2024)
Calculate
Force Timeseries
(Armaoğlu & Monti, 2014)
Model parameters
DP Model
Response timeseries
(Armaoğlu & Monti, 2014)
Final
Statistics
Simulated response
used for training
DeepONet
H_s
Model parameters
If training

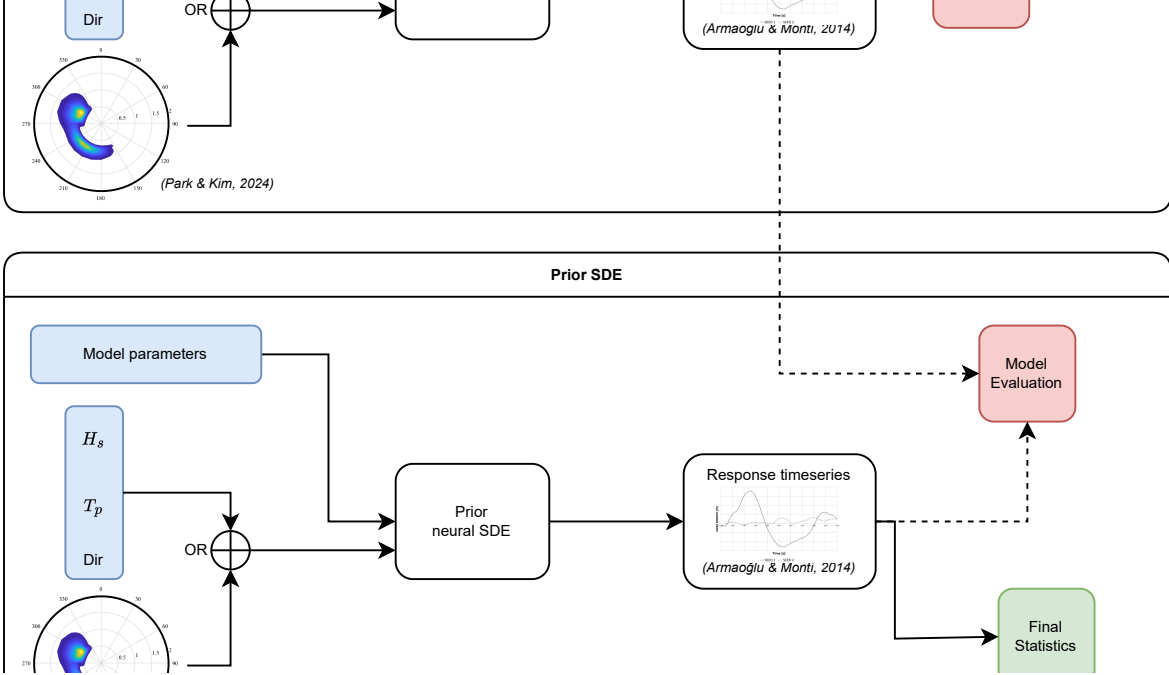

T_p
Dir
OR
(Park & Kim, 2024)
Calculate
Force Timeseries
(Armaoğlu & Monti, 2014)
If training,
use same force
DeepONet
Response timeseries
(Armaoğlu & Monti, 2014)
Final
Statistics

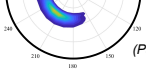

Conventional workflow
H_s
T_p
Dir
OR
(Park & Kim, 2024)
Calculate
Force Timeseries
(Armaoğlu & Monti, 2014)
Model parameters
DP Model
Response timeseries
(Armaoğlu & Monti, 2014)
Final
Statistics


DeepONet
H_s
Model parameters


T_p
Dir
OR
Calculate
Force Timeseries
(Armaoğlu & Monti, 2014)
DeepONet
Response timeseries
(Armaoğlu & Monti, 2014)
Final
Statistics
(Park & Kim, 2024)


Posterior SDE (Only used during training)
Model parameters
H_s
T_p
Posterior
neural SDE
Measurements
(Armaoğlu & Monti, 2014)
Response timeseries
Model
Evaluation


Dir
OR
(Park & Kim, 2024)
(Armaoglu & Monti, 2014)
Prior SDE
Model parameters
H_s
T_p
Dir
OR
Prior
neural SDE
Response timeseries
(Armaoglu & Monti, 2014)
Model
Evaluation
Final
Statistics



(Park & Kim, 2024)